MIL-I-36698A (DSA-DM)
23 June 1975
SUPERSEDING
MIL-I-36698 (DSA-DM)
21 July 1970

MILITARY SPECIFICATION

IMMERSION OIL, MICROSCOPY

This limited coordination Military Specification has been prepared by the Defense Personnel Support Center, based upon currently available technical information, but it has not been approved for promulgation as a coordinated revision of MIL-I-36698 (DSA-DM). However, pending its promulgation as a coordinated Military Specification, it may be used in procurement.

1. SCOPE

1.1 This specification covers a synthetic type oil suitable for use as a medium when conducting the oil immersion method of microscopy.

2. APPLICABLE DOCUMENTS

* 2.1 Specifications and standards. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-566	Boxes, Folding, Paperboard
PPP-B-636	Boxes, Shipping, Fiberboard
PPP-B-676	Boxes, Setup.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

FSC 6640

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(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM Designation D-156	Saybolt Color of Petroleum Products
ASTM Designation D-445	Viscosity of Transparent and Opaque Liquids (Kinematic and Dynamic Viscosities), Test for.
ASTM Designation D-1807	Relative Index and Specific Optical Dispersion of Electrical Insulating Liquids.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Materials. The immersion oil shall consist of mineral and synthetic hydrocarbons and shall contain no polychlorinated biphenyls. The oil shall be non-drying and shall feel sticky to the touch of the finger without showing evidence of polymerization or setting when tested as specified in 4.4.1.

3.2 Refractive index. The immersion oil shall have a refractive index of 1.5150 ± 0.0005 when tested as specified in 4.4.2.

3.3 Viscosity. The immersion oil shall have a viscosity of 1,250 Centistokes ± 10 percent when tested as specified in 4.4.3.

3.4 Color. The immersion oil shall have a Saybolt Color of not less than + 20 when tested as specified in 4.4.4. Immersion oil shall not vary in color within the same lot.

3.5 Label adhesion. The markings on the labels shall remain clear and legible; the labels shall remain firmly fixed to the bottles when tested as specified in 4.4.5.

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3.6 Optical dispersion. The immersion oil shall have an optical dispersion (V-value) of 44 ± 5 when tested as specified in 4.4.6.

* 3.7 Workmanship. Workmanship shall be first class throughout. Immersion oil shall be free of defects listed in Table II and other defects which may detract from its appearance or may impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

* 4.1.2 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.3 Certificate of quality. Certificate of quality, supplied by the manufacturer of the immersion oil, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contract, the contract number, the NSN, the item identification, the name of the material, the lot number, the date of testing, the test method, individual test results and the specification requirements.

4.2 Sampling.

4.2.1 Lot. A lot shall consist of a manufacturer's batch.

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4.2.2 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels shall be as indicated in Table I. Unit of product for examination purposes shall be one 1-fluid ounce bottle of immersion oil.

Table I Sampling for examination

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.0
Minor defects	II	1.5

4.2.3 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and Table III. Unit of product for testing purposes shall be one 1-fluid ounce bottle of immersion oil.

4.3 Examination. Immersion oil shall be examined to determine compliance with all requirements contained in this specification. Examination shall be conducted in accordance with Table II. The immersion oil shall be examined for defects which shall not be restricted to the classified possible defects.

Table II Classification of defects

Categories	Defects
Major	
101	Immersion oil not free from particulate matter.
102	Immersion oil not free from turbidity.
103	Immersion oil varying in color within the lot.
Minor	
201	Bottles not securely capped.

* 4.4 Tests. Tests shall be conducted to determine compliance with specification requirements and in accordance with Table III. Unit of product for test shall be one 1-fluid ounce bottle of immersion oil. Except for label adhesion test, acceptance number shall be zero. Acceptable quality level (percent defective) for label adhesion test shall be 2.5.

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* Table III Sampling for tests

Characteristic	Requirement	Test proccdure	Inspection level
Non-drying	3.1	4.4.1	S-4
Refractive index	3.2	4.4.2	S-4
Viscosity	3.3	4.4.3	S-4
Color	3.4	4.4.4	S-4
Label adhesion	3.5	4.4.5	S-4
Optical dispersion	3.6	4.4.6	S-4

4.4.1 Non-drying. Pipet 5 milliliters (ml.) of immersion oil into a clean, dry evaporating dish and allow it to stand, uncovered, at room temperature for not less than 8 hours. At the end of the exposure time, examine the contents of the dish by touching it with the finger.

4.4.2 Refractive index. Determine the refractive index of the immersion oil by using the Abbé refractometer, or any other refractometer of equal or greater accuracy. The value for the refractive index is to be determined at 23° Centigrade (°C.) in terms of the D line of sodium light.

4.4.3 Viscosity. The viscosity of the immersion oil shall be determined in accordance with ASTM D-445 by using an Ostwald-type viscometer at 23°C.

4.4.4 Color. The color of the immersion oil shall be determined in accordance with ASTM D-156.

4.4.5 Label adhesion. Immerse labeled bottles in water for 4 hours at a temperature of 75° Fahrenheit (°F.) ± 5°. At the end of exposure period, remove bottles and examine labels.

4.4.6 Optical dispersion. The optical dispersion of the immersion oil shall be determined in accordance with ASTM D-1807.

* 4.5 Examination for preparation for delivery. An examination shall be made to determine whether the packaging, packing and marking comply with the requirements of Section 5. Defects shall be scored as specified in Table IV. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2, and the acceptable quality level shall be 4.0 (defects per hundred units).

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Table IV Examination for preparation for delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Packaging and packing materials	Any component missing or damaged.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, tape improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper size, location, sequence or method of application.

* 5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be Level A or C as specified (see 6.1).

5.1.1 Unit of issue. One bottle, containing one fluid ounce as specified, constitutes one unit of issue.

5.1.1.1 Level A.

5.1.1.1.1 Unit package. Immersion oil shall be packaged in a commercially available light-resistant bottle. At the option of the contractor, each bottle may be further packaged in a paperboard box.

5.1.1.1.2 Intermediate package. Twelve units of issue shall be packaged in a box of appropriate size and design, constructed in accordance with PPP-B-566, PPP-B-676 or PPP-B-636, type CF, class domestic. Box design shall include full-height partitions, half-slotted style. The partitions shall form an individual snug-fitting cell for each bottle. Partitions shall be fabricated of material equal to the strength of board used for the fabrication of the box. When individual bottle is packaged in a box, partitions shall not be required. Closure of the intermediate box shall be adequate to prevent accidental opening.

5.1.1.2 Level C. Each unit of issue shall be packaged in standard commercial containers of the size and kind commonly used for the purpose, which will afford the degree of protection required for shipment and use of the end product for its intended purpose.

5.2 Packing. Packing shall be Level A, B or C as specified (see 6.1).

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5.2.1 Level A. One hundred and forty four units of issue shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, class weather-resistant, grade V3c. Box design shall include liner, and top and bottom pads. Liners and pads will not be required when intermediate boxes conform to PPP-B-636. Closure, waterproofing and strapping shall be as specified in the appendix of PPP-B-636.

5.2.2 Level B. One hundred and forty four units of issue shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, type CF, class domestic. Box design shall include liner, and top and bottom pads. Liners and pads will not be required when intermediate boxes conform to PPP-B-636. Closure of box shall be as specified in the appendix of PPP-B-636.

5.2.3 Level C. Immersion oil shall be packed in substantial commercial containers of the type, size and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, to point of delivery called for in the contract or purchase order.

5.3 Marking. Each unit, intermediate and exterior container shall be marked in accordance with MIL-STD-129. Marking shall include the lot (control) number.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. National Stock Number (NSN).
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification covers the following item as listed in the Federal Supply Catalog:

National Stock No.	Item Identification
6640-00-299-9807	IMMERSION OIL, MICROSCOPY, 1 oz.

6.3 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes

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Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

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DOCUMENT IDENTIFIER AND TITLE

MIL-I-36698A (DSA-DM)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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